

How do detect Deepfakes

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Abstract.

This report evaluates if a discriminator taken from a DCGAN-style model with a self-attention layer can be repurposed as a deepfake detector. The discriminator was further trained on a dataset of 140,000 real and fake images. Performance was assessed using accuracy, F1 score, and AUC-ROC across the three train, validation, and test sections of the dataset. The finalised model achieved a 92.92% accuracy, 92.74% F1, and an AUC-ROC of 0.9823 when tested on the testing dataset of 20,000 images (half fake, half real). These results prove that a repurposed discriminator performs very well at detecting deepfake images without overfitting or memorising the data.

1 Introduction

Deepfakes are AI-generated or AI-manipulated images and videos that replace a person's appearance or identity with advanced generative models. Their increasing realism enables applications ranging from entertainment to misinformation, fraud, and privacy violations. Generative Adversarial Networks form the backbone for modern deepfake creation since their adversarial training process enables the synthesis of highly photorealistic human faces. Among these, StyleGAN-based architectures have significantly improved deepfake realism, especially StyleGAN2, providing high-resolution facial synthesis and offering fine-grained control over the attributes. Simultaneously, however, the generative process introduces subtle statistical artifacts. These can be leveraged for deepfake detection; the discriminator part of a GAN inherently learns to identify such artifacts. The report investigates how GANs work, the creation of deepfakes, and how the pretrained GAN discriminator can be repurposed as a deepfake detector. It is structured to first introduce necessary background on the different types of GANs, then describe the experimental method, present results, and discuss implications and limitations. The final section concludes the report and points out directions for future work.

2 Background

Generative Adversarial Networks (GANs) [1] are composed of two parts: one is a generator of synthetic images from random vectors,

and the other is a discriminator, classifying images as real or fake. During training, the generator improves by trying to fool the discriminator, while the latter improves at catching ever more realistic fakes. This adversarial setup lets GANs learn complex visual distributions, especially human faces. Another byproduct of this process is that the discriminator learns characteristic differences between real and generated images, which will be relevant in the context of deepfake detection.

GAN Variants Relevant to Deepfake Generation: DCGAN-2015: The first modern architecture introduced convolutional layers for stable GAN training.

StyleGAN introduced latent space disentanglement and style mixing in 2019.

StyleGAN2-2020: improved the design by fixing artifacts and architectural issues in StyleGAN, improving image sharpness and realism.

StyleGAN3-2021: has improved in geometric consistency and aliasing but it has fewer pretrained models available online in comparison to StyleGAN2.

While highly realistic, there are subtle but detectable inconsistencies in GAN-generated images, including: frequency irregularities boundary artifacts abnormal textures (teeth, eyes, hair) inconsistent lighting It thus learns the differences between them and becomes a practical model for detecting deepfakes [2].

3 Experiments and results

To verify if a GAN discriminator could be used as a deepfake detector, I repurposed and trained the discriminator from a GAN-like convolutional model to distinguish between real and fake images. The model receives 64x64 RGB face images and outputs a probability that the input is real. The dataset consists of 140,000 images in three independent splits: train, validation, and test, each containing separate folders for real and fake images. This guarantees that there would be no image leakage between splits, and hence generalisation was able to be measured faithfully.

All experiments were executed using the CPU version of PyTorch. The final training used default parameters of 5 epochs with early stopping, a batch size of 64, and a learning rate of 0.0001 with the Adam optimizer. Before training the model, some of the images were subject to randomised horizontal flips, colour jittering, and small rotations. This creates realistic variation that is present in real-life images to try to prevent overfitting and improve robustness and versatility of the deepfake detector.

The implementation consisted of two phases. In phase one, a

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GAN was trained from scratch using a DCGAN-inspired architecture, where the discriminator contains one Self-Attention block positioned after the 256-channel convolutional layer. The generator uses transposed convolutional layers with batch normalisation and ReLU activations to progressively upsample a 128-dimensional latent vector into 64 by 64 RGB images, with a self-attention mechanism for long-range spatial dependencies and a final Tanh activation constraining outputs to $[-1, 1]$.



Figure 1: Example output from the generator after 2 epochs

Phase two uses the discriminator from the already trained GAN and trains it further on the 140,000 real and fake images. Although the implementation includes a full GAN (Generator + Discriminator), only the discriminator was employed for testing on real vs. fake images because this component is responsible for identifying real vs fake images.

During the training phase, the model continuously recorded performance history, tracking loss, accuracy, F1 score, and AUC-ROC for both the training and validation sets. At each epoch, the weights resulting in the highest validation accuracy were saved in the `best_detector.pth` file to allow evaluation on unseen test data.

Training behaviour: Training loss decreased gradually across epochs, and training-validation curves converged without evidence of divergence or collapse. Validation AUC and F1 score showed early improvement and a natural plateau.

Test performance:

On the final test set (10,000 real, 10,000 fake):

Loss: 0.1793

Accuracy: 92.92%

Precision: 95.08%

Recall: 90.51%

F1 Score: 92.74%

AUC-ROC: 0.9823

The AUC-ROC remained the most stable metric, indicating that the model learned a strong separation between real and fake images even when prediction thresholds shifted.

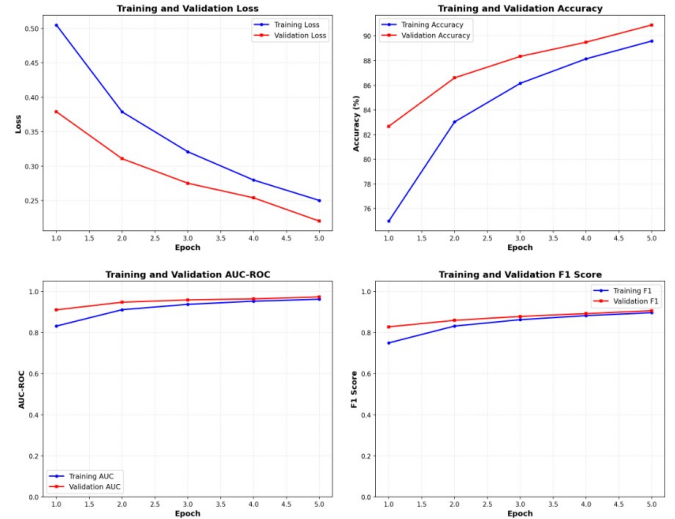


Figure 2: Training and validation curves for loss, accuracy, F1 score, and AUC-ROC

Model Performance: Fake vs Real Classification

	Predicted →	
	Fake	Real
Fake (Actual)	9532 True Negative	468 False Positive
Real (Actual)	949 False Negative	9051 True Positive

Test Results
Loss: 0.1793
Accuracy: 92.92%
Precision: 95.08%
Recall: 90.51%
F1 Score: 92.74%
AUC-ROC: 0.9823

Figure 3: Confusion matrix showing classification outcomes for real and fake images

After testing on 10,000 real images and 10,000 fake images, there were fewer false positives (fake images classified as real) than false negatives (real images classified as fake) this suggests that the model is conservative and strict, which in many sensitive contexts is more desirable for a model to be conservative.

4 Discussion

These experiments demonstrate that a GAN-style discriminator can effectively be repurposed as a deepfake detector when independently retrained on a real/fake dataset. The discriminator demonstrated successful generalisation to unseen data without overfitting, as evidenced through the convergence of accuracy, precision, recall, F1 score, and AUC-ROC metrics. The stability of AUC-ROC across training indicates that meaningful and robust separating features were learned rather than simplistic patterns.

The architectural elements seem to play an important role. Spectral normalisation and self-attention likely helped stabilise training and model longrange spatial dependencies in faces, while dropout reduced the risk of the discriminator memorising training examples. These components combined gave the discriminator a reliable feature extraction pathway for distinguishing real and fake images.

The observed tendency toward fewer false positives compared to false negatives implies that the model adopts a conservative strategy, classifying uncertain examples as fake. In many real-world situations, this may be highly desirable since misclassifying a fake as real can have much worse consequences than the opposite.

Despite such promising results, a number of limitations still persist. The entire model was trained on low-resolution 64 by 64 images; thus, the architecture may have to be changed for the higher-resolution deepfakes normally encountered in practical applications. Moreover, the dataset did not involve real-world artifacts like video compression, motion blur, inconsistent lighting, and temporal inconsistencies—all common features in deepfake media. Performance may consequently be altered in far more complex or adversarial conditions.

Taken together, these findings strongly support the view that GAN discriminators can be repurposed to provide high-accuracy detection of deepfakes, and that even relatively small convolutional architectures can learn discriminative features well when trained under conditions of careful control.

5 Conclusion and future work

This study concludes that a GAN discriminator can be effectively repurposed as a deepfake detector, generalising well, not overfitting, and not memorising the data. Future work will explore training over more epochs and using datasets with bigger variety in images such as compression, lighting variation, and motion blur.

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